

Exercise 3:

Results for function (i):

```
a=0;% lower lim
b=1;% upper lim
analy1 = sqrt(pi)/2;
f1 = @integrand1;

In2 = gauss(f1,a,b,2); % n=2
In3 = gauss(f1,a,b,3); % n=3
In5 = gauss(f1,a,b,5); % n=5
I = [In2 In3 In5]; % vector of the I results
perror1=zeros(1,3);
for i=1:length(perror1)
    perror1(i) = 100*abs((I(i)-analy1)/analy1);
end
fprintf(['Analytical result %f\n' ...
        'Integral n = 2 result: %f \n' ...
        'Integral n = 3 result: %f \n' ...
        'Integral n = 5 result: %f \n' ...
        'For n = 2, the percent error is %f \n' ...
        'For n = 3, the percent error is %f \n' ...
        'For n = 5, the percent error is %f \n'],analy1,In2,In3,In5,perror1(1),perror1(2),perror1(3))
```

```
Analytical result 0.886227
Integral n = 2 result: 0.866989
Integral n = 3 result: 0.876495
Integral n = 5 result: 0.882365
For n = 2, the percent error is 2.170816
For n = 3, the percent error is 1.098075
For n = 5, the percent error is 0.435815
```

For n = 5 we have the minimum percent error (0.43%). That means that the integral from 0 to 1 of

$f(x) = \sqrt{\log\left(\frac{1}{x}\right)}$ using Gauss Quadrature is better approximated using n=5.

Results for exercise (ii):

```
a2=0;% lower lim
b2=pi;% upper lim
analy2 = -(1/2)*pi^2*log(2);
f2 = @integrand2;
I2n2 = gauss(f2,a2,b2,2); % n=2
I2n3 = gauss(f2,a2,b2,3); % n=3
I2n5 = gauss(f2,a2,b2,5); % n=5
I2 = [I2n2 I2n3 I2n5]; % vector of the I results
perror2=zeros(1,3);
for i=1:length(perror2)
    perror2(i) = 100*abs((I2(i)-analy2)/analy2);
end
fprintf(['Analytical result %f \n' ...
        'Integral n = 2 result: %f \n' ...
```

```

'Integral n = 3 result: %f \n' ...
'Integral n = 5 result: %f \n' ...
'For n = 2, the percent error is %f \n' ...
'For n = 3, the percent error is %f \n' ...
'For n = 5, the percent error is %f \n'],analy2,I2n2,I2n3,I2n5,perror2(1),perror2(2),perror

```

```

Analytical result -3.420544
Integral n = 2 result: -2.389427
Integral n = 3 result: -2.904029
Integral n = 5 result: -3.213292
For n = 2, the percent error is 30.144837
For n = 3, the percent error is 15.100394
For n = 5, the percent error is 6.059034

```

The Gauss Quadrature method gave a better result for the integral from 0 to π of $f(x) = x * \log(\sin(x))$ for $n = 5$ with a percent error around 6%.

Results for exercise (ii)

```

a3=0;% lower lim
b3=1;% upper lim
analy3 = pi;
f3 = @integrand3;
I3n2 = gauss(f3,a3,b3,2); % n=2
I3n3 = gauss(f3,a3,b3,3); % n=3
I3n5 = gauss(f3,a3,b3,5); % n=5
I3 = [I3n2 I3n3 I3n5]; % vector of the I results
perror3=zeros(1,3);
for i=1:length(perror3)
    perror3(i) = 100*abs((I3(i)-analy3)/analy3);
end
fprintf(['Analytical result %f \n' ...
'Integral n = 2 result: %f \n' ...
'Integral n = 3 result: %f \n' ...
'Integral n = 5 result: %f \n' ...
'For n = 2, the percent error is %f \n' ...
'For n = 3, the percent error is %f \n' ...
'For n = 5, the percent error is %f \n'],analy3,I3n2,I3n3,I3n5,perror3(1),perror3(2),perror

```

```

Analytical result 3.141593
Integral n = 2 result: 1.794481
Integral n = 3 result: 1.963307
Integral n = 5 result: 2.128654
For n = 2, the percent error is 42.879900
For n = 3, the percent error is 37.506011
For n = 5, the percent error is 32.242848

```

The Gauss Quadrature method gave a better result for the integral from 0 to 1 of $\frac{1}{x(1 - \ln(x)) * \left(\sqrt{\ln\left(\frac{1}{x}\right)}\right)}$

when $n = 5$ with a percent error around 32% which is still a big distance from the actual value of the integral π .

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function [y] = integrand1(x)

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y = sqrt(log(1/x));
end

function [y] = integrand2(x)
y = x*log(sin(x));
end

function [y] = integrand3(x)
w = x*(1-log(x))*sqrt(log(1/x));
y = 1/w;
end

function [I] = gauss(f,a,b,n)
w = zeros(1,n); % weights
zeta = zeros(1,n); % zeta values
    if (n == 1)
        zeta(1)=0;
        w(1)=2;
    elseif (n == 2)
        zeta(1)=1/sqrt(3);
        zeta(2)=-zeta(1);
        w(1)=1;
        w(2)=1;
    elseif (n==3)
        zeta(1)=0;
        zeta(2)=sqrt(3/5);
        zeta(3)=-zeta(2);
        w(1)=8/9; w(2)=(5/9); w(3)=w(2);
    elseif (n == 5)
        zeta(1)=0;
        zeta(2)=sqrt(5-4*sqrt(5/14))/3;
        zeta(3)=-zeta(2);
        zeta(4)=sqrt(5+4*sqrt(5/14))/3;
        zeta(5)=-zeta(4);
        w(1)=128/225;
        w(2)=161/450+13/(180*sqrt(5/14));
        w(3)=w(2);
        w(4)=161/450-13/(180*sqrt(5/14));
        w(5)=w(4);
    else
        disp('This value of n not implemented!')
    end
    I=0;
    alpha=(b-a)/2;
    beta=(b+a)/2;
    for i=1:n
        x=alpha*zeta(i)+beta; % transformation
        I=I+w(i)*f(x); % evaluate our function at x
    end
    I=I*alpha;
end

```